

NO UNRESTRICTED MAHLER PRODUCT FOR THE POLAR-VOLUME VALUATIONS

AGENT LANE 11

STATUS AND EXACT SCOPE

This packet gives a candidate counterexample, likely valid, to the *unrestricted two-sided product* interpretation of the question posed by Mussnig [1]. Within the nonnegative, upper-supported weight class appearing in the source classification, no positive lower bound can hold on all finite super-coercive convex functions. For weights that are nontrivial at positive levels, no finite upper bound can hold even after imposing the strong normalization “even, radial, and minimum zero.”

The source asks broadly about “a similar inequality” for “suitable” functions and convex functions. Therefore this packet does not claim that every more narrowly normalized variant, every one-sided inequality, every different combination of the two quantities, or every weight outside the classified valuation family is impossible.

SOURCE QUESTION

The source is Fabian Mussnig, $SL(n)$ *Invariant Valuations on Super-Coercive Convex Functions*, arXiv:1903.04225, later published in the *Canadian Journal of Mathematics* 73 (2021), 108–130 [1]. The unnumbered remark following Theorem 1.3* appears on PDF page 4. The crop below includes the theorem’s hypothesis on the weight and the complete remark.

Theorem 1.3*. *For $n \geq 2$, a map $Z : \text{Conv}_{\text{sc}}(\mathbb{R}^n, \mathbb{R}) \rightarrow [0, \infty)$ is a continuous, $\text{SL}(n)$ and dually translation invariant valuation if and only if there exist continuous functions $\zeta_0, \zeta_1, \zeta_2 : \mathbb{R} \rightarrow [0, \infty)$ where ζ_1 has finite moment of order $n - 1$ and $\zeta_2(t) = 0$ for all $t \geq T$ with some $T \in \mathbb{R}$ such that*

$$Z(u) = \zeta_0(u(0)) + \int_{\mathbb{R}^n} \zeta_1(u^*(x)) \, dx + \int_{\mathbb{R}^n} \zeta_2(\nabla u(x) \cdot x - u(x)) \, dx \quad (4)$$

for every $u \in \text{Conv}_{\text{sc}}(\mathbb{R}^n, \mathbb{R})$.

Remark. The volume product $V_n(K)V_n(K^*)$ is of significant interest in convex geometric analysis. In particular,

$$c^n V_n(B^n)^2 \leq V_n(K)V_n(K^*) \leq V_n(B^n)^2 \quad (5)$$

for every origin symmetric $K \in \mathcal{K}_{(o)}^n$, i.e. $K = -K$, where B^n denotes the Euclidean unit ball and $c > 0$ is an absolute constant. The right side of (5) is sharp with the maximizers being ellipsoids [37] and is also known as the Blaschke-Santaló inequality [9, 40]. The left side is due to Bourgain and Milman [12] but the optimal constant c is still not known. The famous Mahler conjecture states that the volume product is minimized for affine transforms of cubes (among others) and a proof for the two-dimensional case is due to Mahler [33]. More recently, the conjecture was confirmed for the three-dimensional case [23], but the general case remains open.

Functional versions of (5) for log-concave functions were obtained in [3, 5, 6, 24]. In particular, it was shown that

$$\left(\frac{2\pi}{c}\right)^n \leq \int_{\mathbb{R}^n} \exp(-u(x)) \, dx \int_{\mathbb{R}^n} \exp(-u^*(x)) \, dx \leq (2\pi)^n$$

for suitable convex functions u on $\mathbb{R}^n$, where $c > 0$ is again an absolute constant. Considering Theorem 1.3 and Theorem 1.3*, the question arises if a similar inequality can be obtained using the quantities

$$u \mapsto \int_{\text{dom } u^*} \zeta(\nabla u^*(x) \cdot x - u^*(x)) \, dx \quad \text{and/or} \quad u \mapsto \int_{\text{dom } u} \zeta(\nabla u(x) \cdot x - u(x)) \, dx$$

for suitable functions $\zeta : \mathbb{R} \rightarrow \mathbb{R}$ and convex functions u on $\mathbb{R}^n$.

After recalling the convex-body volume product and a functional Bourgain–Milman/Blaschke–Santaló inequality, the source asks whether a similar inequality can be obtained using

$$u \mapsto \int_{\text{dom } u^*} \zeta(\nabla u^*(y) \cdot y - u^*(y)) \, dy \quad \text{and/or} \quad u \mapsto \int_{\text{dom } u} \zeta(\nabla u(x) \cdot x - u(x)) \, dx$$

for suitable $\zeta : \mathbb{R} \rightarrow \mathbb{R}$ and convex functions u on $\mathbb{R}^n$.

DEFINITIONS

For weights $\zeta, \eta : \mathbb{R} \rightarrow [0, \infty)$ and a finite super-coercive convex function $u : \mathbb{R}^n \rightarrow \mathbb{R}$, set

$$\begin{aligned} \mathcal{A}_\zeta(u) &= \int_{\mathbb{R}^n} \zeta(\nabla u^*(y) \cdot y - u^*(y)) \, dy, \\ \mathcal{B}_\eta(u) &= \int_{\mathbb{R}^n} \eta(\nabla u(x) \cdot x - u(x)) \, dx. \end{aligned}$$

Gradients exist almost everywhere, which is enough for these integrals. We call a weight *upper-supported* if it is continuous and there is $T \in \mathbb{R}$ such that it vanishes for every $t \geq T$. This is exactly the condition imposed on the polar-volume weight ζ_2 in Theorems 1.3 and 1.3* of [1].

PROOF INTUITION

The lower obstruction comes from vertical translation. For the quadratic $u_c(x) = |x|^2/2 + c$, Legendre conjugation changes the sign of c . The argument of ζ in $\mathcal{A}_\zeta$ is $|y|^2/2 + c$. Moving c past the upper-support threshold therefore kills $\mathcal{A}_\zeta$ identically, while the other factor remains finite. Thus a positive Bourgain–Milman-type lower bound cannot survive on the unrestricted class.

The upper obstruction is independent of that translation defect. The normalized radial powers $u_p(x) = |x|^p/p$ have conjugates $u_p^*(y) = |y|^q/q$, where $q = p/(p-1)$. Polar coordinates give exact Mellin moments of the weights. As $p \downarrow 1$ (hence $q \rightarrow \infty$), the first factor decays only like q^{-1} while the second grows like $q^{n/p}$. Their product grows like $q^{n/p-1}$, which diverges in every dimension $n \geq 2$. These witnesses already satisfy origin symmetry and minimum-zero normalization.

THE OBSTRUCTION THEOREM

Theorem 1. *Let $n \geq 2$, and let $\zeta, \eta : \mathbb{R} \rightarrow [0, \infty)$ be continuous upper-supported weights.*

- (i) *On the space of finite super-coercive convex functions,*

$$\inf_u \mathcal{A}_\zeta(u) \mathcal{B}_\eta(u) = 0.$$

In fact, the product is exactly zero for a smooth, strongly convex, origin-symmetric shifted quadratic.

- (ii) *Suppose additionally that each of ζ and η is nonzero at some point of $(0, \infty)$. Even when u is restricted to be finite, C^1 , strictly convex, even, radial, super-coercive, and normalized by $u(0) = \min u = 0$, one has*

$$\sup_u \mathcal{A}_\zeta(u) \mathcal{B}_\eta(u) = +\infty.$$

Proof. Choose thresholds $T_\zeta, T_\eta \in \mathbb{R}$ above which the respective weights vanish.

For (i), let

$$u_c(x) = \frac{|x|^2}{2} + c, \quad u_c^*(y) = \frac{|y|^2}{2} - c.$$

These functions are finite, smooth, strongly convex, super-coercive, and origin symmetric. Direct differentiation gives

$$\nabla u_c^*(y) \cdot y - u_c^*(y) = \frac{|y|^2}{2} + c, \quad \nabla u_c(x) \cdot x - u_c(x) = \frac{|x|^2}{2} - c.$$

If $c \geq T_\zeta$, then the first displayed argument is at least T_ζ everywhere, and consequently

$$\mathcal{A}_\zeta(u_c) = 0.$$

The second factor is finite: its integrand vanishes outside the bounded set $\{|x|^2/2 - c < T_\eta\}$ and is bounded on that set by continuity of η . Thus $\mathcal{A}_\zeta(u_c) \mathcal{B}_\eta(u_c) = 0$. Nonnegativity of the weights proves the claimed infimum.

For (ii), take $p > 1$, put $q = p/(p-1)$, and define

$$u_p(x) = \frac{|x|^p}{p}.$$

Then u_p has all the stated regularity, convexity, symmetry, coercivity, and normalization properties, and

$$u_p^*(y) = \frac{|y|^q}{q}.$$

The two arguments in the valuations are

$$\nabla u_p^*(y) \cdot y - u_p^*(y) = \frac{|y|^q}{p}, \quad \nabla u_p(x) \cdot x - u_p(x) = \frac{|x|^p}{q}.$$

Write $\kappa_n = \text{vol}(B^n)$. Polar coordinates followed by $t = r^q/p$ in the first integral and $t = r^p/q$ in the second give the exact formulas

$$(1) \quad \mathcal{A}_\zeta(u_p) = \frac{n\kappa_n}{q} p^{n/q} \int_0^\infty \zeta(t) t^{n/q-1} dt,$$

$$(2) \quad \mathcal{B}_\eta(u_p) = \frac{n\kappa_n}{p} q^{n/p} \int_0^\infty \eta(t) t^{n/p-1} dt.$$

Both integrals are finite: on $[0, \infty)$ each weight is bounded and has compact support, while both exponents are positive.

By the extra nontriviality assumption and continuity, there are compact intervals $I_\zeta, I_\eta \subset (0, \infty)$ on which the corresponding weights are bounded below by positive constants. As $p \downarrow 1$, the exponents n/q and n/p stay in bounded positive ranges (the first tends to 0 and the second to n). Restriction of the moment integrals in (1)–(2) to those fixed compact intervals therefore gives constants $c_1, c_2 > 0$, independent of p sufficiently close to 1, such that

$$\mathcal{A}_\zeta(u_p) \geq \frac{c_1}{q}, \quad \mathcal{B}_\eta(u_p) \geq c_2 q^{n/p}.$$

Hence

$$\mathcal{A}_\zeta(u_p) \mathcal{B}_\eta(u_p) \geq c_1 c_2 q^{n/p-1} \longrightarrow +\infty,$$

because $n/p - 1 \rightarrow n - 1 > 0$. This proves (ii). $\square$

Corollary 2. *For every pair of nonnegative upper-supported valuation-class weights, there is no positive constant L such that*

$$L \leq \mathcal{A}_\zeta(u) \mathcal{B}_\eta(u)$$

for every finite super-coercive convex u . In particular, no two-sided positive/finite Mahler–Santaló product inequality for the two displayed quantities can hold on that unrestricted function class. If a single weight ζ is nonzero somewhere on $(0, \infty)$, then the same-weight product $\mathcal{A}_\zeta(u) \mathcal{B}_\zeta(u)$ also admits no finite universal upper bound, even on the normalized class in Theorem 1(ii).

Proof. The lower statement is Theorem 1(i); the upper statement is part (ii) with $\eta = \zeta$. $\square$

VERIFICATION NOTES

The proof is analytic and does not depend on computation. The reusable script

`code/verify_radial_families.py`

uses the sample weight

$$\zeta(t) = (1 - t)_+.$$

For this weight,

$$\int_0^\infty \zeta(t) t^{a-1} dt = \int_0^1 (1 - t) t^{a-1} dt = \frac{1}{a(a+1)}.$$

The script compares this exact moment with numerical quadrature, evaluates (1)–(2) for $n \in \{2, 3\}$ and six values of p between 1.5 and 1.01, and checks the shifted-quadratic zero. Run from this packet directory with

`python code/verify_radial_families.py.`

The output confirms the formulas and rapid product growth. It is a sanity check only; the exact substitutions in the proof establish the theorem.

NOVELTY CHECK AND LIMITATIONS

Before promotion, the run’s four lightweight indexes were searched:

```
registry_index.tsv  solutions/index.tsv
attempts/index.tsv  proof_gaps/index.tsv.
```

The terms included arXiv:1903.04225, the exact title, **polar volume valuation**, **Mahler product**, and the displayed integrands. No duplicate or answer packet was found.

Bounded arXiv/web searches on 2026-07-23 used the arXiv id, the exact source question, the displayed gradient integrands, **polar volume valuation inequality**, **Hessian valuation volume product**, and author/title combinations. The search found the source, Mussnig’s earlier classification of volume and polar volume [2], and the later paper of Mussnig and Ulivelli [3]. The latter studies Brunn–Minkowski-type and isoperimetric inequalities for functional intrinsic volumes, not this Legendre-dual product: inspection of its full arXiv text found neither the phrase “polar volume” nor arXiv:1903.04225. No direct resolution of the product question was found. Novelty remains candidate-level pending expert bibliographic review.

The exact source sentence leaves “similar inequality,” “suitable functions,” and “and/or” unspecified. The theorem completely rules out a positive-lower/finite-upper product inequality on the unrestricted source class for its nonnegative upper-supported weights. It does not rule out:

- a one-sided inequality whose direction survives the examples;
- a different algebraic combination of $\mathcal{A}_\zeta$ and $\mathcal{B}_\eta$;
- weights outside the upper-supported valuation class;
- a narrower function class or normalization excluding both vertical shifts and the radial-power degeneration.

HUMAN-REVIEW RECOMMENDATION

The mathematical review focus is short: verify the two Legendre transforms, the two radial substitutions, and the uniform lower bounds on the fixed positive intervals. The more important review question is interpretive: whether the source’s intended “similar inequality” includes the natural two-sided product formalized here. Subject to that scope judgment, the counterexample is complete.

REFERENCES

- [1] F. Mussnig, *SL(n) invariant valuations on super-coercive convex functions*, Canad. J. Math. **73** (2021), 108–130; arXiv:1903.04225.
- [2] F. Mussnig, *Volume, polar volume and Euler characteristic for convex functions*, Adv. Math. **344** (2019), 340–373; arXiv:1806.11084.
- [3] F. Mussnig and J. Ulivelli, *Inequalities and counterexamples for functional intrinsic volumes and beyond*, J. London Math. Soc. **113** (2026), no. 1, Art. e70422; arXiv:2412.05001.